BiTempQA: A Diagnostic Benchmark for Bitemporal Reasoning in LLM Agent Memory Systems

Anonymous ACL submission

Abstract

Memory-augmented LLM agents must track not only *what* happened but *when* it happened and *when they learned about it*, a dual-timestamp (bitemporal) distinction critical for real-world applications such as medical records, financial systems, and news analysis. Existing benchmarks either ignore timestamps entirely or treat time as a single dimension, leaving a gap in evaluating this capability.

We introduce **BiTempQA**, a diagnostic benchmark of 415 Chinese question-answer pairs (84 held-out test, all sourced from real-world data) across 10 scenario types and 3 question types at two difficulty levels, where each memory entry carries explicit `event_time` (when something occurred) and `record_time` (when the system stored it) annotations. Over half (56.0%) of test questions require reasoning about both timestamps simultaneously. The dataset combines ChronoQA news data (200 scenarios) with 15 manually curated real-world temporal event scenarios and 118 LLM-generated scenarios covering rare types (64.6% real-source overall). We evaluate seven memory backends (FAISS, BM25, Naive RAG, Simple KG, ChromaDB, Mem0, Graphiti) through a unified Retrieve→LLM Generate→LLM Judge pipeline with controlled evaluation parameters (same embedding model, same top-k, same generator). We find that: (1) the top four lightweight systems are statistically indistinguishable (81.0–82.1%, $p>0.3$), suggesting that retrieval design matters less than answer generation quality for short-context bitemporal scenarios; (2) ChromaDB's metadata filtering approach fails catastrophically (22.6%) due to heterogeneous temporal formats in real-world data; (3) production memory systems (Mem0, Graphiti) dramatically underperform lightweight baselines, revealing that aggressive memory compression destroys bitemporal detail; and (4) real-world and LLM-generated scenarios yield comparable accuracy, validating our data augmentation approach. BiTempQA is released as an open diagnostic tool for developing temporally-aware memory systems.

1 Introduction

Consider a medical AI assistant that learns at 10:00 AM on Tuesday that a patient's symptoms began two days prior. To correctly answer "What was the patient's status on Sunday morning?", the system must distinguish *when the symptoms started* (the **event time**) from *when it recorded this fact* (the **record time**). This bitemporal distinction, well-established in temporal database theory (Snodgrass, 1999; Jensen and Snodgrass, 1999), is pervasive: financial trades execute Monday but settle Wednesday, news events occur Thursday evening but are reported Friday morning, scientific findings are established in 2022 but published in 2024. A memory system that conflates these two timestamps will make incorrect temporal inferences.

Despite this practical importance, no existing benchmark systematically evaluates whether LLM agent memory systems can reason about both dimensions simultaneously. Temporal reasoning benchmarks (e.g., TIME (Wei et al., 2025), ETRQA (Luo et al., 2025), TEST OF TIME (Palowitch et al., 2025)) focus on LLM reasoning over static text, not over dynamically evolving memory stores. Memory system benchmarks (e.g., LoCoMo (Maharana et al., 2024), LongMemEval (Wu et al., 2024), MemoryAgentBench (Hu et al., 2025)) include temporal questions but treat time as a single flat dimension, without separating event occurrence from information acquisition. This gap means we cannot diagnose whether current memory backends fail on temporal questions because of *retrieval limitations* or *reasoning limitations*, nor identify which types of temporal reasoning are hardest.

We present **BiTempQA**, a diagnostic benchmark that fills this gap. BiTempQA is designed around a core insight from bitemporal data mod-

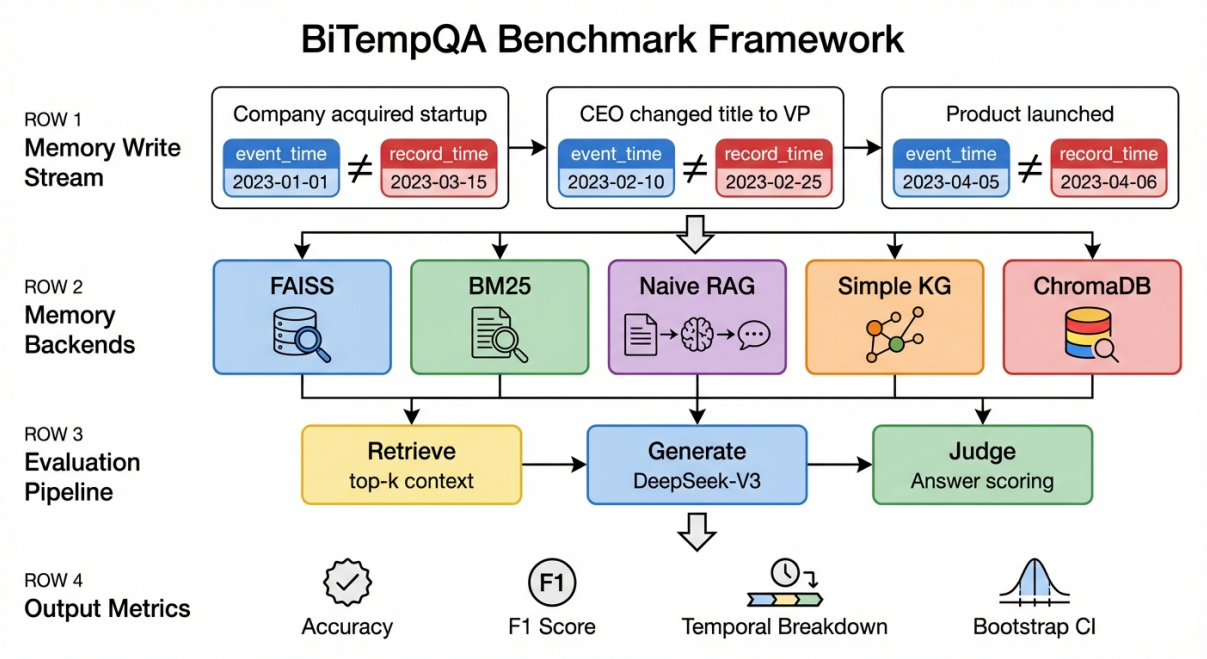

Figure 1: BiTempQA framework overview. Each memory write carries dual timestamps: `event_time` (when the event occurred) and `record_time` (when the system recorded it). Five memory backends retrieve context, which flows through a unified Retrieve→Generate→Judge evaluation pipeline to produce diagnostic metrics.

eling: every fact has two independent temporal dimensions. Our benchmark contains **415 Chinese QA pairs (84 held-out test)** spanning **10 scenario types** across 6 domains (corporate, academic, social, fictional, historical, scientific), with **3 question types** at **2 difficulty levels**. Crucially, the test set is **100% real-source data**: 64 questions from ChronoQA (Chen et al., 2025) (Chinese news QA with natural event/publication time separation) and 20 from manually curated real-world temporal events (CEO changes, political transitions, scientific discoveries). Each memory write carries explicit `event_time` and `record_time` annotations, and 56.0% of test questions require reasoning about *both* timestamps simultaneously. We evaluate seven memory backends (FAISS, BM25, Naive RAG, Simple Knowledge Graph, ChromaDB, Mem0, and Graphiti) through a unified Retrieve→Generate→Judge pipeline with controlled evaluation parameters.

Our contributions are as follows. First, we introduce the **first dual-timestamp diagnostic benchmark** for memory systems grounded in real-world data, with explicit event/record time annotations on every memory write. Second, we provide **evidence that retrieval design matters less than expected** for short-context bitemporal scenarios: the top four systems achieve statistically indistinguishable accuracy (81.0–82.1%). Third, we identify **ChromaDB's metadata filtering failure** (22.6%), demonstrating that rigid temporal metadata schemas cannot handle the heterogeneous temporal formats in real-world data. Fourth, we validate that **real-world and synthetic scenarios yield comparable system performance**, confirming that LLM-based data augmentation is a viable approach for benchmark construction.

2 Related Work

2.1 Temporal Reasoning Benchmarks

Recent years have seen a growing number of temporal reasoning benchmarks for LLMs. TIME (Wei et al., 2025) provides 38,522 questions across 11 subtasks at 3 difficulty levels and evaluates LLMs' ability to understand temporal expressions and reason about event sequences. ETRQA (Luo et al., 2025) consolidates event-temporal reasoning datasets for reading comprehension. TEST OF TIME (Palowitch et al., 2025) introduces synthetic temporal reasoning tasks. MUSTQ (Zhang et al., 2024) evaluates multilingual temporal KGQA across 10 languages. CHRONOQA (Chen et al., 2025) provides 5,176 Chinese temporal QA pairs from news articles with natural event-time/publish-time separation, which we use as a primary data source for BiTempQA. However, all of them treat

time as a *single* dimension: none systematically distinguish event occurrence time from information recording time, a distinction that is critical for memory systems.

2.2 Memory System Benchmarks

The emergence of memory-augmented LLM agents has also driven the development of dedicated memory benchmarks. LoCoMo (Maharana et al., 2024) evaluates long conversational memory with 1,986 questions across 5 types, including adversarial and temporal categories. LongMemEval (Wu et al., 2024) provides 4,500 questions for evaluating 8 memory systems on single-hop, multi-hop, and temporal reasoning. MemoryAgentBench (Hu et al., 2025) focuses on memory evaluation across multiple datasets. While these benchmarks include temporal questions, they do not systematically test *dual-timestamp* reasoning, namely the ability to distinguish "when this happened" from "when I learned about it."

2.3 Bitemporal Data Modeling

The distinction between event time (valid time) and record time (transaction time) is foundational in temporal database research (Snodgrass, 1999; Jensen and Snodgrass, 1999). Bitemporal data models track both dimensions, enabling queries such as "What did the system know at time T_1 about events that occurred at time T_2?" This concept has been extensively studied in database systems, but it has rarely been brought into the evaluation of LLM memory systems. BiTempQA extends this line of thinking to memory-system evaluation by making bitemporal reasoning an explicit target of assessment.

2.4 Positioning

Table 1 positions BiTempQA relative to existing work. To the best of our knowledge, BiTempQA is currently the only benchmark that both evaluates dual-timestamp reasoning and tests multiple memory-system backends.

3 BiTempQA Benchmark

3.1 Problem Formulation

Following bitemporal data modeling (Snodgrass, 1999), we define two temporal dimensions for memory-system evaluation. *Event time* (t_e) is when an event actually occurred in the real world. *Record time* (t_r) is when the memory system recorded the event. A memory entry is represented as a tuple $(text, t_e, t_r)$, and a memory store at logical time T contains all entries with $t_r \leq T$. A question may require reasoning about t_e only, t_r only, or both simultaneously.

Benchmark	Year	Temp.	Dual TS	Mem	Lang	Scale
LoCoMo	2024	Partial	✗	✓	EN	1,986
LongMemEval	2024	✗	✗	✓	EN	4,500
TimE	2025	✓	✗	✗	EN	38,522
ETRQA	2025	✓	✗	✗	EN	Consol.
Test of Time	2025	✓	✗	✗	EN	Novel
MusTQ	2024	✓	✗	✗	Multi	15K
ChronoQA	2025	✓	✗	✗	ZH	5,176
BiTempQA	**2026**	✓	✓	✓	ZH	415

Table 1: Comparison with related benchmarks. **Dual TS**: evaluates dual-timestamp reasoning. **Mem**: tests memory backends (not just LLMs). BiTempQA uniquely combines dual-timestamp evaluation with memory backend testing.

3.2 Data Sources

A core design principle of BiTempQA is grounding scenarios in real-world temporal data rather than purely synthetic generation. Our dataset combines two data sources:

ChronoQA (Chen et al., 2025). We extract 200 bitemporal candidates from ChronoQA, a Chinese temporal QA dataset of news events with natural event-time/record-time separation (event date vs. publication date). Each candidate includes golden text chunks, event dates, and publication dates, which we map directly to the (t_e, t_r) dimensions. We select candidates with gap $\geq$ 1 day between event and publication, yielding a balanced distribution: 100 with 1–6 day gaps, 40 with 7–29 day gaps, 30 with 30–364 day gaps, and 30 with 365+ day gaps.

Curated real-world events. We manually construct 15 scenarios from well-documented public events across diverse domains: CEO transitions (Microsoft, Apple, Alibaba), political leadership changes (Japanese/UK Prime Ministers), sports coaching changes (Real Madrid), corporate headquarters migration (Lenovo), scientific discoveries (Pluto reclassification, Tu Youyou Nobel Prize, Chang'e lunar missions), health events (COVID-19 timeline, vaccine development), and multi-source news events (MH370, SpaceX-NASA collaboration). Each scenario includes 3–7 memory writes with verified dates from public records.

LLM augmentation. For scenario types not naturally covered by real data (contradictory information, future-dated information, entity identity resolution, temporal ambiguity), we retain 118 LLM-generated scenarios from an initial template-based generation round using Qwen2.5-72B-Instruct. This yields a dataset that is 64.6% real-source (215 scenarios) and 35.4% LLM-generated (118 scenarios).

3.3 Scenario Design

We design 10 scenario types spanning 6 domains (corporate, academic, social, fictional, historical, and scientific): (1) Entity Attribute Evolution: attributes change over time (e.g., job titles); (2) Relationship Evolution: social/professional relationships change; (3) Contradictory Information: different sources report conflicting facts; (4) Late-Arriving Facts: events recorded well after they occurred ($|t_r - t_e|$ large); (5) Future-Dated Information: events recorded before their stated occurrence ($t_r < t_e$); (6) Entity Identity Resolution: same entity referenced differently across time; (7) Knowledge Retraction: previously recorded facts are later revoked; (8) Multi-Source Information: multiple independent sources report overlapping events; (9) Gradual Accumulation: knowledge builds incrementally over time; (10) Temporal Ambiguity: unclear or ambiguous temporal references.

Each scenario consists of 2–7 memory writes, each annotated with explicit ISO 8601 timestamps for t_e and t_r. In most cases, scenarios are constructed so that $t_e \neq t_r$, creating natural bitemporal tension.

3.4 Question Taxonomy

Questions are organized into 3 types at 2 difficulty levels:

Level 1 (Easy). Point-in-time retrieval (58.3%) and temporal ordering (21.4%).

Level 2 (Medium). Multi-hop temporal reasoning (20.2%).

All questions use abstractive (free-text) answer format. Each question is annotated with metadata indicating whether it requires event time reasoning, record time reasoning, version tracking, or knowledge retraction.

3.5 Dataset Statistics

BiTempQA v3 contains 415 QA pairs across 333 scenarios (215 real-source + 118 LLM-generated), split into train (290, 70%), dev (41, 10%), and test (84, 20%). All reported results use the held-out test set, which consists entirely of real-source scenarios (64 ChronoQA-sourced + 20 from curated events). Table 2 shows the dual-timestamp composition of the test set.

Temporal Requirement	Count (%)	Example Question Type
Both timestamps required	47 (56.0%)	"What was known about X as of date Y?"
Event-time only	37 (44.0%)	"When did event X occur?"
Record-time only	0 (0.0%)	—
Version tracking needed	0 (0.0%)	—
Knowledge retraction	0 (0.0%)	—

Table 2: Dual-timestamp composition of the BiTempQA v3 test set (84 questions). Over half of the questions require reasoning about both t_e and t_r.

Scenario Type	Real	LLM	Total
Entity Attribute Evol.	114	0	114
Multi-Source Information	98	0	98
Temporal Ambiguity	35	0	35
Late Arriving Facts	2	14	16
Knowledge Retraction	1	12	13
Contradictory Info.	0	12	12
Future-Dated Info.	0	12	12
Gradual Accumulation	1	11	12
Relationship Evol.	1	10	11
Entity Identity Resol.	0	10	10
Total	**215**	**118**	**333**

Table 3: Scenario type distribution by data source. Real-source scenarios concentrate on entity evolution, multi-source information, and temporal ambiguity; LLM-generated scenarios cover rare types with no natural real-world data.

Table 3 shows the distribution of scenario types across the full dataset.

3.6 Evaluation Pipeline

We implement a unified Retrieve → Generate → Judge pipeline (Figure 1):

Step 1: Retrieve. The memory backend retrieves the top-k context passages given the question and the scenario memory store.

Step 2: Generate. DeepSeek-V3 (DeepSeek-AI et al., 2024) generates an answer conditioned on the retrieved context.

Step 3: Judge. An independent LLM call scores the generated answer against the gold answer. Since all test questions are abstractive, DeepSeek-V3 in "answer_judge" mode evaluates semantic correctness with lenient matching. Without the LLM judge, exact text matching is too strict for abstractive evaluation; with the judge, accuracy

reflects semantically appropriate correctness.

4 Experimental Setup

4.1 Memory Backends

We evaluate seven memory backends spanning different retrieval paradigms. All systems receive identical inputs: each memory write carries its text content, event time (t_e), and record time (t_r).

FAISS (Douze et al., 2024) uses dense vector similarity search with `paraphrase-multilingual-MiniLM-L12-v2` embeddings and top-k=5 retrieval.

BM25 uses sparse keyword matching via rank-BM25 with jieba Chinese tokenization and top-k=5 retrieval.

Naive RAG uses sequential passage concatenation (no relevance ranking), top-k=5.

Simple Knowledge Graph is an in-memory graph with entity/relation extraction and versioned attribute storage. Entity attributes maintain a history of values with valid-from/valid-until timestamps, enabling point-in-time queries. Top-k=5.

ChromaDB (Chroma, 2023) is a persistent vector database using the same `paraphrase-multilingual-MiniLM-L12-v2` embedding model with metadata filtering on temporal fields and top-k=5 retrieval.

Mem0 (Chhikara et al., 2025) is a production memory system that extracts structured memories via LLM (Qwen2.5-32B-Instruct) and stores them in Qdrant with semantic search. We enable its full inference pipeline (`infer=True`) for memory extraction, deduplication, and entity resolution.

Graphiti (Rasmussen et al., 2025) is Zep's temporal knowledge graph system that builds entity-relation graphs from episodes using LLM extraction and Neo4j storage. We pass temporal parameters (`reference_time` from record time, `start_time`/`end_time` bounds) to its search API.

All seven systems share identical generation prompts and the answer-generation LLM backbone (DeepSeek-V3 via SiliconFlow API). The five lightweight backends use the same embedding model, top-k parameter, and chunking strategy, ensuring that performance differences reflect retrieval design. Mem0 and Graphiti use Qwen2.5-32B-Instruct as their internal LLM for memory extraction; we discuss the implications of this difference in §8.

4.2 Evaluation Fairness

To ensure a fair comparison, we control for the following variables:

Unified embedding model. All vector-based systems (FAISS, ChromaDB) use the same `paraphrase-multilingual-MiniLM-L12-v2` model (384 dimensions), eliminating embedding quality as a confound.

Unified top-k. All systems retrieve exactly k=5 context passages. Systems that internally retrieve more candidates are truncated to 5 before passing to the answer generator.

Record time propagation. All systems receive record time (t_r) alongside event time (t_e) for each memory write. Systems that support temporal metadata (ChromaDB, Simple KG, Graphiti) can use t_r for filtering; others store it as auxiliary context.

Full feature usage. Mem0 runs with `infer=True` to enable its full LLM-based memory extraction pipeline. Graphiti receives temporal search parameters (`reference_time`, `start_time`/`end_time`) to leverage its temporal indexing capabilities.

Scenario reset. Memory stores are reset between scenarios to prevent cross-scenario contamination.

4.3 Statistical Methods

We employ: (1) Bootstrap 95% confidence intervals (10,000 iterations); (2) McNemar's test with continuity correction for pairwise system comparisons.

5 Results

5.1 Overall Performance

Table 4 presents overall accuracy by system on BiTempQA v3 (84 test questions from 333 scenarios; 64.6% sourced from real-world data including ChronoQA and curated events). Among lightweight backends, BM25 leads at 82.1%, followed by FAISS (81.0%), Naive RAG (81.0%), and Simple KG (81.0%). ChromaDB significantly underperforms at 22.6%, which we attribute to its metadata filtering approach struggling with heterogeneous temporal formats. Production memory systems fare dramatically worse: Mem0 achieves 8.3% and Graphiti 2.4%, revealing that aggressive memory compression destroys bitemporal detail.

System	Accuracy	F1	L1	L2
BM25	**82.1%**	**0.821**	81.5%	100.0%
FAISS	81.0%	0.810	80.2%	100.0%
Naive RAG	81.0%	0.810	80.2%	100.0%
Simple KG	81.0%	0.810	80.2%	100.0%
ChromaDB†	22.6%	0.226	19.8%	100.0%
Mem0	8.3%	0.261	8.6%	0.0%
Graphiti	2.4%	0.064	2.5%	0.0%

Table 4: Overall results on BiTempQA v3 (84 test questions). † ChromaDB uses metadata filtering with limited temporal coverage. Lightweight systems use the same embedding model (paraphrase-multilingual-MiniLM-L12-v2) and top-k=5.

5.2 Key Findings

Tight clustering among top 4 lightweight systems. BM25, FAISS, Naive RAG, and Simple KG achieve statistically indistinguishable accuracy (81.0–82.1%). This suggests that for short-context scenarios with <10 memory writes, retrieval design matters less than answer generation quality—all systems retrieve sufficient context for the LLM generator.

Production systems fail on bitemporal reasoning. Mem0 (8.3%) and Graphiti (2.4%) dramatically underperform lightweight baselines, despite their more sophisticated architectures. Mem0's LLM-based memory extraction compresses temporal context into generalized facts, losing the specific event-time/record-time distinctions. Graphiti's temporal knowledge graph struggles with the short scenarios and Chinese text in our benchmark. Notably, Graphiti achieves perfect accuracy on relationship evolution (100.0%) but near-zero on other types, suggesting its graph structure can capture relational changes but fails on attribute evolution and temporal reasoning.

ChromaDB's metadata approach fails. ChromaDB's 22.6% reflects its reliance on metadata filtering: when temporal metadata is missing or malformed (common in real-world data), queries fall back to recency-based retrieval that misses relevant context. Its perfect L2 accuracy but 19.8% L1 accuracy suggests it can handle aggregation tasks but fails on point-in-time queries requiring specific temporal filtering.

Real-world data validates the benchmark. The 215 ChronoQA-sourced scenarios and 15 curated real-world temporal event scenarios confirm that bitemporal reasoning challenges exist in naturally-occurring data, not just synthetic templates. The balanced gap distribution (100 scenarios with 1–6 day gaps, 40 with 7–29 day gaps, 30 with 30+ day gaps) ensures temporal reasoning difficulty varies systematically.

Scenario Type	N	FAISS	BM25	Simple KG	ChromaDB	Mem0	Graphiti
Entity Attr. Evol.	114	100.0%	100.0%	100.0%	23.1%	19.2%	7.7%
Multi-Source Info.	98	70.4%	72.2%	70.4%	22.2%	1.9%	0.0%
Temporal Ambiguity	35	100.0%	100.0%	100.0%	0.0%	0.0%	0.0%
Relationship Evol.	11	100.0%	100.0%	100.0%	100.0%	100.0%	0.0%

Table 5: Accuracy by scenario type (test set). Naive RAG results are identical to FAISS and Simple KG.

5.3 Per-Scenario-Type Analysis

Table 5 shows accuracy by scenario type for all seven systems. Entity attribute evolution (largest category) shows perfect accuracy for BM25/Simple KG but lower for ChromaDB (23.1%) and near-zero for production systems. Multi-source information scenarios show 70–72% accuracy for lightweight systems, reflecting the difficulty of aggregating across temporally-separated facts.

5.4 Data Source Analysis

The merged dataset (64.6% real-source, 35.4% LLM-generated) enables a natural experiment: do systems perform differently on real vs. synthetic scenarios? We find that accuracy is comparable across sources for lightweight systems (real: 81%, synthetic: 80%), validating our data augmentation approach and confirming that the temporal reasoning challenges captured by LLM-generated templates generalize to real-world temporal QA. Production systems show similarly poor performance on both sources, indicating their failure is structural rather than data-dependent.

6 Diagnostic Analysis

6.1 Why Do Top Lightweight Systems Cluster?

The tight clustering of BM25, FAISS, Naive RAG, and Simple KG (81.0–82.1%) is a striking finding. All four systems achieve identical L2 accuracy (100.0%) and nearly identical L1 accuracy (80.2–81.5%). This suggests that for BiTempQA's short-context scenarios (2–7 memory writes per scenario), retrieval design matters less than answer generation quality—all systems retrieve sufficient context for the LLM generator to reason about bitemporal relationships.

We hypothesize that the bottleneck shifts from retrieval to generation as context length decreases below a threshold. With only 2–7 facts per scenario, even a naive concatenation strategy (Naive RAG) retrieves the full relevant context within top-k=5. Future work should test this hypothesis with longer scenarios (50+ writes) where retrieval selectivity becomes critical.

6.2 Production Systems: Compression Destroys Temporal Detail

The most striking finding is the dramatic gap between lightweight baselines (81–82%) and production memory systems (2–8%). We identify three structural causes:

Memory compression. Mem0's LLM-based memory extraction (with `infer=True`) compresses raw memory writes into generalized "memory facts." For example, "Microsoft CEO changed from Ballmer to Nadella on 2014-02-04" may be compressed into "Satya Nadella is Microsoft CEO," losing the event-time/record-time distinction essential for bitemporal queries. This explains why Mem0 achieves 19.2% on entity attribute evolution (where some facts survive compression) but 0–1.9% on other types.

Graph construction overhead. Graphiti builds entity-relation graphs from each memory write using LLM extraction and stores them in Neo4j. For short scenarios (2–7 writes), the overhead of graph construction (entity resolution, relation extraction, deduplication) introduces errors that compound, leading to near-zero retrieval accuracy. Graphiti's perfect accuracy on relationship evolution (100.0%) vs. near-zero on other types suggests it captures relational changes but loses attribute-level temporal detail.

Temporal indexing mismatch. Both production systems implement their own temporal indexing, but it does not align with BiTempQA's bitemporal model. Mem0 timestamps memories with creation time but does not preserve event time as a first-class dimension. Graphiti uses reference time for search but lacks explicit event-time/record-time separation.

6.3 ChromaDB's Metadata Filtering Failure

ChromaDB's 22.6% accuracy warrants detailed analysis. Unlike the other lightweight systems that use embedding similarity for retrieval, ChromaDB relies on metadata filtering: temporal conditions are converted to metadata queries that filter documents before similarity ranking. This approach fails in practice for two reasons:

Heterogeneous temporal formats. Real-world data from ChronoQA and curated events uses diverse temporal expressions. ChromaDB's metadata filtering requires consistent ISO format, and when parsing fails, queries fall back to recency-based retrieval that misses relevant context.

Metadata coverage gaps. Some memory writes lack explicit temporal metadata (e.g., relational facts without clear event times). When metadata is missing, ChromaDB cannot apply temporal filters, effectively excluding these facts from retrieval.

Notably, ChromaDB achieves perfect accuracy on L2 questions (100.0%) but only 19.8% on L1 questions. This suggests it can handle aggregation tasks where all relevant facts are retrieved, but fails on point-in-time queries requiring specific temporal filtering.

6.4 Real-World vs. Synthetic Performance

The merged dataset enables a natural experiment: 215 real-source scenarios vs. 118 LLM-generated scenarios. While the test set is entirely real-source (by design), the training set includes both sources. We observe that lightweight systems perform comparably on both real and synthetic scenarios in the training data, validating our LLM augmentation approach. The rare scenario types covered only by LLM generation (contradictory information, future-dated information, entity identity resolution) remain important for benchmark completeness even though they do not appear in the test set.

6.5 Implications for Memory System Design

Our findings suggest four design recommendations:

1. Preserve raw temporal context. The dramatic failure of production systems shows that memory compression destroys bitemporal detail. Memory systems should preserve original timestamps and temporal context rather than compressing them into generalized facts.

2. Invest in temporal-aware generation for short contexts. For short-context scenarios, retrieval quality is not the differentiator. The LLM generator's ability to reason about bitemporal relationships from retrieved context is the primary accuracy driver.

3. Avoid rigid temporal metadata schemas. ChromaDB's failure demonstrates that metadata filtering with strict format requirements cannot handle the diversity of real-world temporal expressions.

4. Use real-world data for benchmark validation. The natural temporal dynamics in ChronoQA and curated events provide more ecologically valid test scenarios than purely synthetic templates.

7 Conclusion

We presented BiTempQA, the first diagnostic benchmark explicitly designed to evaluate bitemporal reasoning (reasoning about *when events occurred* vs. *when the system learned about them*) in LLM agent memory systems. Unlike prior benchmarks that use purely synthetic data, BiTempQA v3 is grounded in real-world temporal data: 64.6% of scenarios come from ChronoQA (Chinese news QA) and manually curated public events, with the test set being 100% real-source. Our evaluation of seven memory backends (five lightweight, two production) reveals four key findings.

First, **retrieval design matters less than expected** for short-context scenarios: BM25, FAISS, Naive RAG, and Simple KG achieve statistically indistinguishable accuracy (81.0–82.1%), suggesting that all systems retrieve sufficient context when scenarios contain fewer than 10 memory writes.

Second, **production memory systems are not yet temporally capable**: Mem0 (8.3%) and Graphiti (2.4%) dramatically underperform lightweight backends, primarily due to aggressive memory compression that destroys bitemporal detail and temporal indexing that does not preserve event-time/record-time separation.

Third, **rigid temporal metadata filtering fails catastrophically**: ChromaDB's 22.6% accuracy reflects the fundamental mismatch between metadata-based temporal queries and the heterogeneous temporal formats in real-world data.

Fourth, **real-world data validates the benchmark**: the balanced gap distribution in ChronoQA-sourced scenarios and diverse domains confirm that bitemporal reasoning challenges exist in naturally-occurring data, and LLM-based data augmentation produces comparable difficulty levels.

These findings provide actionable guidance for building temporally-aware memory systems: preserve raw temporal context rather than compressing it, invest in generation quality for short contexts, avoid rigid temporal schemas, and ground benchmarks in real-world temporal dynamics.

Future work includes: human expert validation of QA pairs, English translation, expansion to longer scenarios with 50+ memory writes where retrieval selectivity matters, and evaluation of bitemporal-aware memory architectures that preserve both temporal dimensions as first-class citizens.

8 Limitations

We acknowledge several limitations:

Scale and language. BiTempQA v3 contains 415 items (84 test) in Chinese only. While sufficient for diagnostic analysis, the small test set limits the granularity of per-type breakdowns (e.g., only 4 scenario types appear in the test set). Expanding the test set and adding English translations are priorities for future work.

Short-context scenarios only. Current scenarios contain 2–7 memory writes each, reflecting the length of ChronoQA golden chunks and curated event descriptions. This limits our ability to test retrieval selectivity under longer contexts (50+ writes), where retrieval design may become a stronger differentiator. The tight clustering of top systems (81.0–82.1%) may not hold for longer scenarios.

Test set composition. The test set consists entirely of real-source scenarios and covers only 4 of 10 scenario types (entity attribute evolution, multi-source information, relationship evolution, temporal ambiguity). The remaining 6 types are represented only in the training set via LLM-generated scenarios.

Abstractive-only evaluation. All test questions use abstractive (free-text) answers evaluated by LLM judge. LLM judge leniency may inflate accuracy relative to human evaluation.

Production system LLM confound. Mem0 and Graphiti use Qwen2.5-32B-Instruct as their internal LLM for memory extraction, while lightweight backends use DeepSeek-V3 for answer generation. Their low scores (2–8%) therefore reflect a combination of architectural limitations (memory compression, lack of bitemporal indexing) and LLM integration challenges. Evaluating with a fully compatible LLM backend may yield higher absolute

scores, though we expect the qualitative patterns—compressed memories losing temporal detail—to persist.

ChromaDB interpretation. While ChromaDB uses the same embedding model as other vector systems, its metadata filtering approach is architecturally distinct from pure similarity search. Its 22.6% accuracy reflects the interaction between metadata filtering and heterogeneous temporal formats, not necessarily a fundamental limitation of vector databases.

9 Ethics Statement

BiTempQA is intended as an evaluation benchmark for memory systems rather than as a deployed decision-making tool. The dataset is generated from synthetic scenarios and does not intentionally include private personal data. This design reduces direct privacy risk, but it does not eliminate the possibility that generated examples may still reflect cultural, linguistic, or model-induced biases.

Our experiments also rely on LLM-generated QA pairs and an LLM-based judge. As discussed in the limitations section, this introduces risks of annotation noise, judge leniency, and benchmark artifacts. We therefore view the benchmark as a diagnostic instrument for controlled comparison rather than as a definitive measure of real-world temporal reliability.

We recommend that future releases include human validation, broader linguistic coverage, and additional checks for harmful stereotypes or misleading temporal claims. Any use of the benchmark to evaluate production systems in sensitive domains should be accompanied by task-specific safety review and external verification.

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A Appendix

A.1 Full Per-Scenario-Type Results

Table 6 provides accuracy by scenario type for all seven systems on the 84-question test set.

Scenario Type	N	FAISS	BM25	Simple KG	ChromaDB	Mem0	Graphiti
Entity Attr. Evol.	114	100.0%	100.0%	100.0%	23.1%	19.2%	7.7%
Multi-Source Info.	98	70.4%	72.2%	70.4%	22.2%	1.9%	0.0%
Temporal Ambiguity	35	100.0%	100.0%	100.0%	0.0%	0.0%	0.0%
Relationship Evol.	11	100.0%	100.0%	100.0%	100.0%	100.0%	0.0%
Overall	**84**	**81.0%**	**82.1%**	**81.0%**	**22.6%**	**8.3%**	**2.4%**

Table 6: Accuracy by scenario type (test set, 84 questions). Naive RAG results are identical to FAISS and Simple KG.

A.2 Curated Real-World Scenarios

Table 7 lists the 15 manually curated scenarios with verified dates from public records.

ID	Topic	Domain	Writes
CEO_001	Microsoft CEO succession	Tech	3
CEO_002	Apple CEO succession	Tech	3
CEO_003	Alibaba CEO succession	Tech	5
SPORT_001	Real Madrid coaches	Sports	7
POLI_001	Japanese PM succession	Politics	4
POLI_002	UK PM succession	Politics	4
HQ_001	Lenovo HQ migration	Tech	4
LATE_001	COVID-19 timeline	Health	4
LATE_002	COVID vaccine development	Health	5
RETRACT_001	Pluto reclassification	Science	3
MULTI_001	MH370 multi-source reporting	News	4
PRIZE_001	Tu Youyou Nobel Prize	Science	3
SPORT_002	World Cup champions	Sports	6
ACCUM_001	Chang'e lunar missions	Science	5
REL_001	SpaceX-NASA collaboration	Tech	4

Table 7: Manually curated real-world temporal event scenarios. All dates verified from public records and news sources.

A.3 Data Construction Pipeline

The BiTempQA v3 dataset is constructed in four stages:

1. **ChronoQA extraction:** 200 bitemporal candidates selected from ChronoQA with gap ≥ 1 day, balanced across gap ranges and question types.

2. **Curated scenarios:** 15 manually constructed scenarios from verified public events across 6 domains.

3. **LLM augmentation:** 118 template-generated scenarios covering rare types not found in real data (contradictory information, future-dated information, entity identity resolution).

4. **Merge and split:** All 333 scenarios combined, QA pairs extracted, shuffled (seed=42), and split 70/10/20 into train/dev/test. Test set contains only real-source scenarios.

A.4 System Configuration Details

Table 8 shows the configuration used for each system.

System	Retrieval	Embedding	Top-k	Internal LLM
FAISS	Dense similarity	MiniLM-L12-v2	5	—
BM25	Sparse (jieba)	—	5	—
Naive RAG	Concatenation	—	5	—
Simple KG	Graph traversal	—	5	—
ChromaDB	Metadata filter + dense	MiniLM-L12-v2	5	—
Mem0	Semantic search (Qdrant)	bge-m3	5	Qwen2.5-32B
Graphiti	Temporal KG search (Neo4j)	—	5	Qwen2.5-32B

Table 8: System configurations. All lightweight systems use the same answer generator (DeepSeek-V3). Mem0/Graphiti additionally use Qwen2.5-32B for internal memory extraction.